



SUPERIOR MOTION SENSOR Using Microwave Doppler Radar

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An earlier design introduced a touch solid-state AC switch. The new version advances this concept with a sophisticated human radar sensing system that can also be configured for IoT and smart automation. The system uses 70 per cent Indian components and is entirely Indian-engineered.

At its core lies a microwave doppler radar sensor that detects human motion with far greater

reliability than traditional PIR or IR-based systems. Unlike PIR sensors that depend on heat signatures and often malfunction under sunlight, near heaters, or behind curtains, the microwave radar delivers superior penetration through non-metallic barriers, immunity to heat-induced false triggers, reduced sensitivity to small animals, and omnidirectional detection without requiring a direct line of sight.

Where most motion switches rely on PIR sensors and frequently cause false triggers, this radar-based system eliminates such issues. It also integrates an LDR to function only under low-light or nighttime conditions, making it ideal for automated lighting systems. Fig. 1 illustrates the prototype design.

The components for the system have been selected such that Indian substitutes are available wherever possible, enabling over 70 per cent localisation in the design. Table 1 lists the Bill of Materials, including part names, numbers, manufacturers, and Indian substitute options where applicable.

Design

In this upgraded version, the separate optocoupler and triac ICs have been replaced with a single integrated optotriac that combines both functions. This component, sourced from an Indian manufacturer, can directly

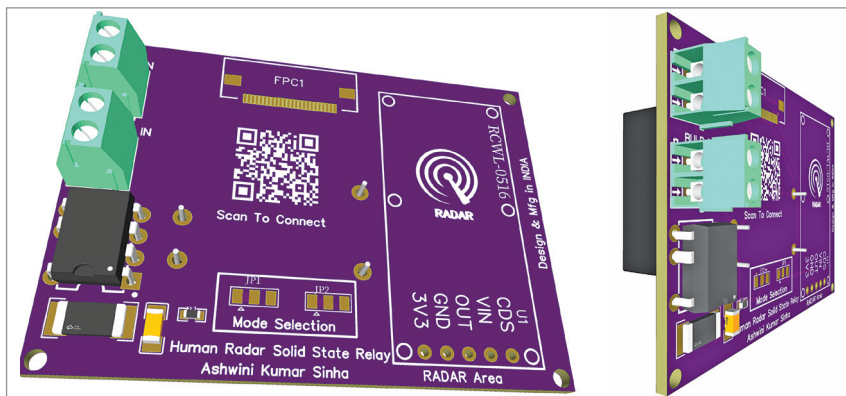


Fig. 1: Author's prototype design

TABLE 1: BILL OF MATERIALS

Name	Designator	Footprint	QTY	Man. Part	Manufacturer	Indian Substitute Part No.	Indian Manufacturer
100nF	C3	C1206	1	CC1206KKX7RYBB104	YAGEO	1206B104K251CT	Walsin
DB301V-3.5-2P-GN	CON1, CON2	CONN-TH_DB301V-3.5-2P-GN	2	DB301V-3.5-2P-GN-S	DORABO	TL001R-3.5-2P	Connectwell
FPC Connector	FPC1	OV2640 FPC 24P	1	AFC01-S24FCA-00	JUSHUO	ZF5S-24-01-T-WT	Samtec Electronics India
VO2223A-X007T	IC1	DIP-8_7P-L9.7-W6.4-P2.54-LS7.6-BL-PE7	1	VO2223A-X007T	VISHAY	VO2223A-X007T	VISHAY
120Ω	R1	R0603	1	0603WAF1200T5E	UNI-ROYAL	CRCW0603120R-FKEA	VISHAY
100Ω	R3	RES-SMD_L6.4-W3.2-R2512	1	SR2512FK-7W100RL	YAGEO	100Ω SMD resistor	HTR-INDIA, Watts Electronics, Kusum/KOHM
RCWL 0516	U1	RCWL-0516	1	RCWL-0516		NA	NA
LDR	U2	UNIVIBE LAMP-LDR ARRANGEMENT_LDR3	1			NA	NA
LD05-23B05R2_C42432360	U4	PWRM-TH_LD05-23BXXR2	1	LD05-23B05R2	LZTEC	RECOM RAC05-05SK	RECOM Power India Pvt Ltd
IoT Board	MOD1	NA		ESP S3, Arduino IoT Kit		IndusBoard Coin V2	IndusBoard V2